

Plausible Prior Attempts

Respiratory therapists treat patients with lung and heart problems. They can perform rehabilitation on patients and manage the patients airways. The therapist's responsibilities include but are not limited to; managing the ventilator, assessing lung capacity, performing diagnostic tests, developing and implementing treatment plans, monitor and recording progress of said treatment plan, and finally educating patients about diseases and their available treatments.

Pulmonary Function Testing is the main way these therapists can assess how well your lungs work. It can determine how much air goes into and out of your lungs, how much air goes from your lungs to your blood and how well your lungs work during exercise. These Pulmonary Function tests (PFTs) are used by medical professionals to assess lung health and function. They are used to determine the causes of issues with the function of lungs, eventually leading to diagnosing certain diseases. These tests are non-invasive and measure how well your lungs inhale, exhale, and exchange oxygen and carbon dioxide.

Spirometry is an easy and accessible way to measure basic lung functions. It gives the clinician a basic understanding of the patient's lung with general information on their respiratory system. This is where **plethysmography** comes in: *"A patient may appear to have restrictive disease, but the clinician will be sure by doing plethysmography," Burns said, noting that, in some cases, a patient could have both restrictive and obstructive patterns. "Spirometry and plethysmography together would verify this. By knowing this you can maximize patient treatment. Plethysmography is used to confirm a restrictive component of lung disease—that spirometry can only suggest—and is likely one of the most important functions of body plethysmography. While gas dilution methods can measure the total lung capacity, body plethysmography is often considered to be more accurate and often more repeatable."* People need to be able to get diagnosed to minimize the effects of COPD. Spirometry is used on people with symptoms but people who are asymptomatic can have COPD. It's important to diagnose and test. Spirometry can tell if there is something wrong with your breathing and its degree.

Plethysmography determines how much air is in your lungs after you take a deep breath. The machine is a box that you enter and the machine takes measurements. It can detect a specific disease.

Radar is popular in detecting chest movements in infants from reflected signals. Radars send out signals and it analyzes how they are reflected back from the infant's body. These are very precise ways to measure and are non-contact, which can improve the convenience of this method. External interference like with unexpected infant movements can affect the precision and accuracy of the method.

Thermography can track temperature changes in areas like the nose in order to monitor breathing patterns. Thermography is able to track respiratory flow as well as motion.

Chest Measurement by Hand is one of the most commonly used methods by physical therapists, in which they measure the chest circumference at the peak of exhalation and at the peak of inhalation. While this may be effective in adults, the method becomes more difficult to use with younger and younger patients.

Analysis of Prior Attempts

Chest Measurement by Hand

[Measuring Thoracic Excursion: Reliability of the Cloth Tape Measure Technique - Journal of Osteopathic Medicine](#)

The task of establishing normative data for chest expansion is difficult in that the data available is limited and typically does not help in any way. Dr. Brossman wishes to change this by establishing a baseline for normative chest expansion data that can be used in research, teaching, and PT assessments. Dr. Brossman has discussed how her current method for measuring chest expansion is to use a sewing measurement tape and wrap it around the body of the patient. She has stressed that this method has more negative than positive sides.

Strengths

The advantage of using a tape measure is that it allows for data to be gathered, albeit possibly inaccurate, about chest expansion. This can assist in characterizing different illnesses and physical issues in the lungs with certain ranges of chest expansion. As stated in the Journal of Osteopathic Medicine study titled "Measuring Thoracic Excursion: Reliability of the Cloth Tape Measure Technique - Journal of Osteopathic Medicine", the tape measure is found to be highly reliable in adult males.

This finding, however, cannot be generalized to all age groups as needs vastly differ and so does handling

Weaknesses

The negatives heavily outweigh the positives with this method, as Dr. Brossman discussed the level of difficulty this method brings to any PT assessor. To be able to successfully measure chest expansion using a tape measure, a high level of physical therapy expertise is required as there are many factors that can result in inaccurate measurements. One such factor is if the tape measure is not held correctly, it will drape down the back of the patient and show incorrect measurements as the patient breathes. Dr. Brossman also wishes to expand past the scope of this method, as this method can obtain measurements during one time frame, but cannot monitor long term in order to find normative data.

NeoBeat

Comparing pulse rate measurement in newborns using conventional and dry-electrode ECG monitors - PMC

This device is used to monitor heart rate in newborn babies. This device uses dry stainless-steel electrodes embedded in the spring-elastic buckle which is placed over the newborn's chest or abdomen to measure the electrical conductance.

Strengths

NeoBeat has advantages to its use such as it is easy to use and quick to display a heart rate measure. This could be useful to our own project in incorporating a real time heart rate measurement of the infant's heart. For this device the slowest reading time was 2.5 seconds We can include this in our own device by including a way that realtime readings can be cataloged and seen.

The NeoBeat shows accurate heart rate reports in comparison to the gold industry standard. According to the article *Comparing pulse rate measurement in newborns using conventional and dry-electrode ECG monitors*, NeoBeat when compared to other devices that are considered the golden standard such as the conventional ECG and PO is far superior in measurement in heart rate. It was observed by Vonderer et al., an underestimation of pulse rate in the first minutes after birth was observed in approximately 61% of patients (Twists). Additionally the article states that, "The PPVs indicate that nearly all HRs <100 reported by PO are false-positive. The moderate PPV of NeoBeat originates in a single patient where for a handful of time instances NeoBeat reports HR <100 and ECG ≥100 bpm" (Twist). For the majority of these instances, NeoBeat remains within the reliability margin, as the difference with ECG is <10 bpm. Overall NeoBeat remains to have a better performance compared to the performance of Pulse Oximetry (PO).

The dry electrodes capture ECG-based signal readings with a simplified and streamlined process for obtaining the heart rate of the infant. Additionally, Twists reports “Neobeat is able to monitor heart rate in temperatures up to 40 degrees Celsius and/or up to 90% humidity” (Twist).

NeoBeat can be used in a variation of settings such as high resource and low resource setting. Meaning this device can be used in an environment with minimal hospital equipment and a lot of hospital equipment.

In application to our own project our team can apply the use of dry electrodes to take accurate heart measurement. Our team should also include a way to display the heart rate of the infant either on a screen or application in real time. Additionally, our team should also consider including the versatility of the environments the Baby breathing monitoring device can be used. We should take inspiration from the Neobeat and make it solid in that the Baby breathing device can be operated in both high and low equipment environments. As well as, including humidity and temperature tolerance as a factor to enhance performance.

Weaknesses

The Neobeat for all its strengths will have its drawbacks. In the article *Perceptions and experiences of Congolese midwives implementing a low-cost battery-operated heart rate meter during newborn resuscitation*, the testimonies of midwives in the usefulness of the NeoBeat in their practice report that “Challenges of use of NeoBeat included cleaning, charging and insufficient quantity of devices”(Thornton).

For our own device it should be considered to make it easy to clean and a sufficient way to charge the device. Overall the main objective of this device is to measure chest circumference, not heart rate. Dr. Broseman has mentioned that having the information for heart rate would be nice, but not absolutely necessary to have.

Contactless Radar-Based Breathing Monitoring

[What are the specific characteristics of a baby's skin? — Typology](#)

[The Growing Child: Newborn | Johns Hopkins Medicine](#)

Procedure

This device is to assist infants that are born premature in breathing. Due to their immature organ system these infants need extra assistance in breathing. In order to ensure infant survival a non-invasive method has been developed to monitor the breathing and vitals of infants. The radar works as an ultrawideband that takes breathing estimates for the infant.

Strengths and Weaknesses:

This device is to assist infants that are born premature in breathing. Due to their immature organ system these infants need extra assistance in breathing. In order to ensure infant survival a non-invasive method has been developed to monitor the breathing and vitals of infants. The radar works as an ultrawideband that takes breathing estimates for the infant. According to the article *Contactless radar-based breathing monitoring of premature infants in the neonatal intensive care unit*, “ radar signals can penetrate through different materials (such as plexiglass, clothing, mattresses and blankets), and are not affected by skin pigmentation or ambient light levels”(Beltrao).

The advantages of this device is that it reduces hygiene risks and does not cause discomfort, skin irritation or skin damage. According to the article *The specific characteristics of a baby's skin*, “A newborn's skin absorbs external compounds such as pollution particles, bacteria, and allergens much more easily than an adult's skin, making it particularly vulnerable”(Pauline). Furthermore this is why it is such a benefit that there is less skin damage and contact to the infant when the reading for the infant's breathing is taking place. According to the article *Contactless radar-based breathing monitoring of premature infants in the neonatal intensive care unit*, “ radar signals can penetrate through different materials (such as plexiglass, clothing, mattresses and blankets), and are not affected by skin pigmentation or ambient light levels”(Beltrao). There is no additional contact with the child beyond having them placed in a containment area.

For our own design this device would follow the design requirements to have proper hygiene as infants are vulnerable to outside pathogens. This device for the most part is contactless and minimizes the amount of germs the child is coming into contact with.

Another strength of this device is that it does not restrain the patient in their movement. This itself is very important for infants. Newborn children and infants need to move to develop their muscles and coordination. As the infant's nervous system starts to develop into reflexes to stem their development.

Weakness

A weakness of this device is that the radar because of reduced transmitting power the signal can be buried in the background noise, or masked by stronger external interference, such as movements of infants. According to the article *Contactless radar-based breathing monitoring of premature infants in the neonatal intensive care unit*, "Most research on contactless vital sign monitoring with radar sensors focus on a single-person setup under ideal motionless conditions. In practical monitoring situations, the subject may often move body parts like hands, legs or torso, and even the entire body. These unwanted but unavoidable movements are usually called random body movements (RBM)"(Beltrao). This can inhibit readings, but can be overcome with specific signal processing methods to aid in having reliable readings, but can potentially raise the price of this piece of technology.

For the Baby breathing device it would be valuable to include having minimal contact with the infant to promote hygiene and to protect the infant's skin from pathogens when readings would be taken. Additionally, having the device being non-constricted to allow movement for the infant will also be something to consider. As well as ensuring our group's baby breathing device can indeed get accurate readings without assistance will be something to include. The Radar breathing device can give unreliable readings that can be interfered with, but can be helped with additional technology. It's important for our device to function on its own without assistance to get reliable readings and decrease the cost of this device.

Thermography

Thermography has been explored as a non-contact technique to monitor and assess chest wall motion and tidal volume, even though it primarily measures surface temperature rather than physical function and movement. Investigations showed that thermal imaging could represent temperature sudden changes in the thorax during breathing cycles. This suggests potential use for monitoring respiratory patterns (Jones & Plassmann 2002). Other studies used high-resolution thermal cameras to track temperature gradients on the chest. This allowed for estimations of respiratory rates without physical sensors (Murthy et al 2014). Recent work applied thermography to map chest wall perfusion post-cardiac surgery, which indirectly reflected regional expansion. The studies proved how thermography can be a viable attempt for contactless respiratory monitoring. They also emphasized how this method provides qualitative data, however, rather than quantitative, which is the goal.

Strengths: The non-contact and non-invasive feature makes this idea for monitoring respiration in patients where traditional methods aren't as easy to use. These can be infants but also people like burn victims or post-surgical patients (Murphy et al 2014). Since it relies on radiation that is naturally emitted into the body, it gets rid of the need for contact with skin. This can be useful when on infants because it reduces discomfort, contamination, and interference with natural breathing patterns. Modern infrared cameras also provide high spatial and temporal resolution, which lets us visualize subtle thermal changes across the chest instead of one spot (Pereira et al 2022). Thermography also allows for the monitoring of multiple regions of the chest at the same time, which is a key factor in monitoring chest expansion. It offers a more in depth view of the symmetry compared to one point sensors.

Weaknesses: Thermography presents limitations that restrict its effectiveness. The main drawback is that the imaging measures surface temperature and not physical movement. This means that it can only indirectly measure chest motion and not really provide precise measurements of expansion (Murphy et al 2014). The accuracy is also very dependent on environmental conditions. This can cause variability in the surface temperature patterns and ultimately affect the measurements (Jones & Plassmann 2022). Skin emissivity differences and the presence of clothing can also affect the signal leading to inconsistent results. Another limitation is the lack of standardized calibration that connects temperature changes to actual thoracic displacement. This hinders comparison across other studies and ultimately limits its reliability.

Belts/ Wearable Technologies

Historically, chest belts have been used for measuring resistance-band sensors. Each change in resistance values, caused by the expansion and contraction of the chest wall during breathing, results in a detectable change in the current registered by the sensor.

According to the article *Advances in Respiratory Monitoring: A Comprehensive Review of Wearable and Remote Technologies*, "Chest straps are primarily designed for recording RR or RV and are known for their accuracy in measuring these parameters compared to alternative wearables" (Vitazkova). One example is the Airgo belt, which uses a resistance band system placed on the ribcage that detects chest circumference. It is composed of stretchable materials with a silver coated yarn, uses a microprocessor for data collection, Bluetooth for wireless communication, and inertial measurement for motion detection.

These Chest straps are precise. They can offer great stability, especially during physical movements and can also continuously monitor throughout the day. While they are monitoring, it gives real time respiration data.

The advantages of this device is that it takes accurate measurements for recording respiration rate and respiration volumes. This device is efficient in being able to be placed over clothing and can be worn during physical activities. According to the article, "Chest straps excel in accuracy and stability, so they are ideal for continuous monitoring. They are preferred for use in sports and fitness, where stability and precision are crucial even in movement" (Vitazkova). This device can proceed to do continuous monitoring through the day giving real time data during the respiration monitoring. This device is also cost effective.

This can be applied into our own device by including that our baby monitoring device could give real time readings and function throughout the day and well as being worn during physical movement. This would be something important to consider because babies typically have a major inclination to move around. Having this monitoring device be able to be worn and allow the child to move will help reduce any stress the infant may feel. Another thing to consider is getting all day monitoring of the child. Including this in our design would allow for more data to be collected to monitor the infant throughout the day.

The weaknesses of this device is that it can be uncomfortable for some people. Additionally, the sensors are susceptible to outside interferences. The article states, "these sensors can be easily interfered with by environmental factors such as humidity or temperature, which can cause inaccurate results"(Vitazkova). Outside interferences can impair readings and cause the data to be less accurate.

Something to consider for our design is to address the fact of outside factors interfering with readings, such as humidity and skin conditions. This is important because without these factors getting addressed it will inhibit the reliability of the readings. Our device will need to address this issue.

Impulse Oscillometry

["Current place of impulse oscillometry in the assessment of pulmonary diseases." - ScienceDirect](#)

[Impulse oscillometry: The state-of-art for lung function testing - PMC](#)

[Lung Function Assessment by Impulse Oscillometry in Adults - PMC.](#)

Analysis

The Impulse oscillometry measures the respiratory system's response to the flow of sound waves at specific frequencies.

Patient's Role

A patient must sit still on the equipment and breath normally through the mouthpiece for around 30 seconds. A nose peg or clamp will be used to isolate the breathing only from the mouth. The patient must be able to follow directions as instructed.

Procedure

After the child is seated with the head in the neutral position, the nose is clipped, and the cheeks and lower jaw are supported to prevent the shunting of oscillations into the upper airway. The patient takes tidal breathing through a tight mouthpiece in a closed system. The loudspeaker delivers multiple impulses over a period. The minimum acquisition time without artefacts is 16 s in children <12 y of age and 3 s in adults [15]. IOS should be performed before performing forced expiratory manoeuvres, such as spirometry, if both tests are done in the same session [15].

Strengths and Weaknesses:

[Lung Function Tests in Infants and Children - PMC](#)

[Impulse oscillometry](#)

[Small vulnerable newborns: the urgent need of strong actions in Peru and the entire Latin America](#)

This method's weaknesses are that it requires patient cooperation and physical capacity. According to the article, *Impulse oscillometry: The state-of-art for lung function testing*, "forceful expiratory and inspiratory maneuvers of spirometry require patient cooperation and physical capacity that is usually lacking in young children below 4 years of age" (Desiraju). This could be particularly difficult to make a child cooperate to sit still. There are no uniform standards to facilitate a set of unified results and development of a standardized method for calculating values based on measured parameters. It can create inconsistent readings and test performance. It is also hard to compare the readings between this machine and different equipment. As the usual amount of times for the reading to take place is 3 times.

Some things to consider when designing our own device is to ensure child cooperation is at a minimum that it will not be difficult for a patient to get tested. A consideration for our own device is that the design should be that an infant should be physically co-operate with the machine monitoring.

As well as developing a standard method for calculating values to get readings that have actual meaning to the data that is being collected for the child's breathing.

The main advantages of FOT/IOS It is also a great alternative for spirometry, it is noninvasive, and it also requires minimal patient cooperation. In the article *Lung Function Assessment by Impulse Oscillometry in Adults*, it is noted that “The advantage of oscillometry, being non-invasive and also effort-independent”(Suppini). It is quick and easy and can be performed in children from 2–3 y of age to the elderly [19–21]. It can also be used where spirometry is contraindicated, such as in recent surgery, recurrent pneumothorax, spirometry-related bronchospasm and during pandemics where forceful blowing is discouraged.

These strengths to consider adding to our own device is to make it non-invasive so that the child does not feel distress or become hurt. Babies are very fragile and are prone to getting sick easily. As according to the article, *Small vulnerable newborns: the urgent need of strong actions in Peru and the entire Latin America*, “ More than 25% of all newborns worldwide—are born too soon” (Huicho).This can be especially useful for premature infants because this can often leads them vulnerable to diseases or having other medical complications that result in death.

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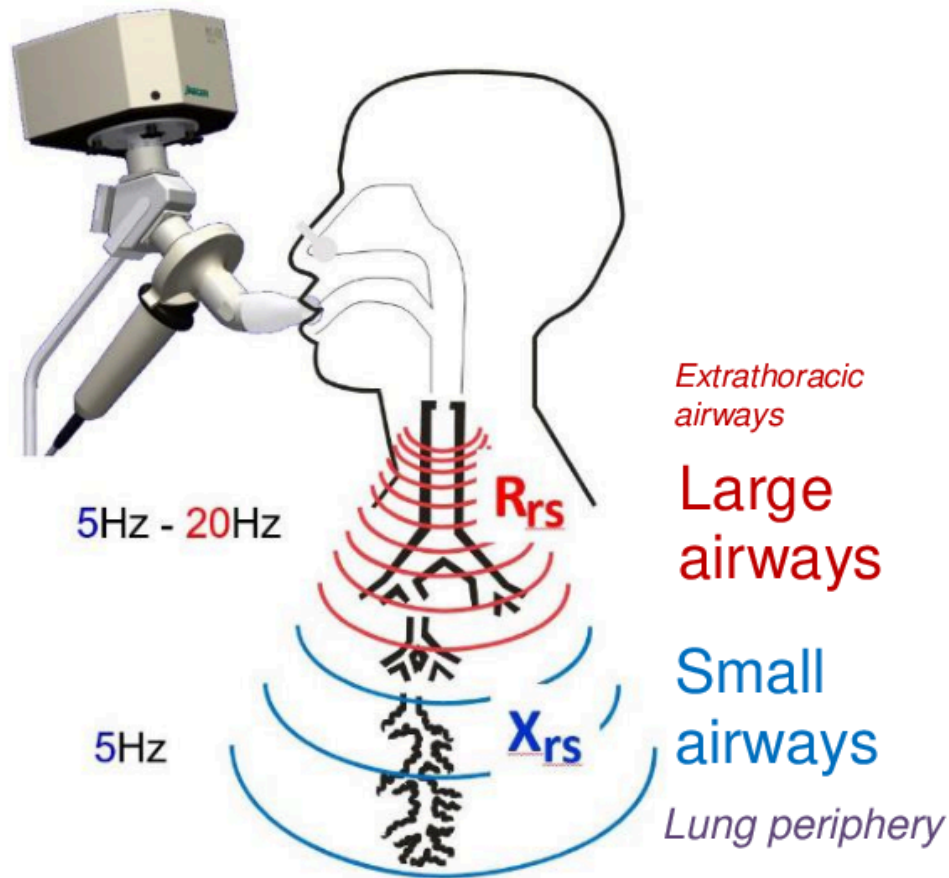


Fig. 2

Spirometry

[Spirometry - Mayo Clinic](#)

[Lung Function Tests in Infants and Children - PMC](#)

[A comparison of alternative selection methods for reporting spirometric parameters in healthy adults](#)

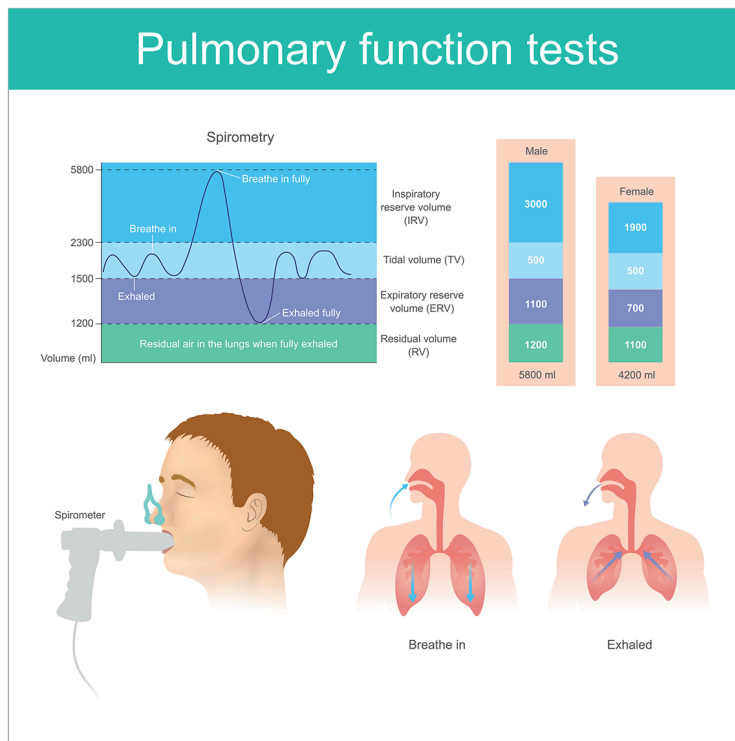
Analysis

Spirometry is a pulmonary function test (PFT) that measures how much air you breathe in, how much you breathe out, and how quickly you breathe out. Spirometry is often used as an initial test performed for patients with respiratory symptoms and used to characterize the severity of disease, optimize therapy, and monitor lung function. The test provides the patient's Forced vital capacity (FVC), ("the largest amount of air that you can forcefully breathe out after breathing in as deeply as you can") and their forced expiratory volume (FEV) ("how much air you can force out of your lungs in one second"). The FEV reading helps "healthcare professionals figure out how serious your

breathing problem is. Lower FEV-1 readings mean greater blockages in the bronchial tubes.”

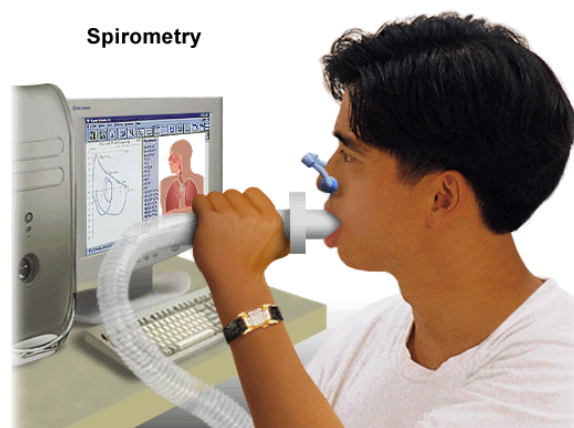
This test is the most used and available PFT. Height, weight, age, sex, ethnicity, environmental factors, patient cooperation and effort, and technical factors all play a role in the effectiveness of spirometry.

Figure 1



Patient Role

The test requires the patient to exhale in a tube attached to the Spirometer. During the test a clip is placed on the patient's nose. The patient will take a deep breath in and out for a small period of time in the tube. The following image showcases the spirometer in use.



Performing a Spirometry Test

After recording height, weight, sex and ethnicity, the patient is seated on a chair or may stand. There are four steps of the forced vital capacity (FVC) manoeuvre:

1. Rapid and maximal inspiration (breathe in) with lips tightly around the mouthpiece
2. A “blast” of expiration (breathe out)
3. continued blowing of air fast for 6 s in children (although it may be 3 s in younger children) and finally
4. complete inspiration (taking in the maximum possible breath)

The last inspiration step is optional and only required if inspiratory parameters are required

Strengths and weaknesses:

Spirometry is one of the most commonly used methods to assess lung function so it is highly accessible assuming the patient has access to a healthcare facility.

According to the article *Lung Function Tests in Infants and Children*, “Lung function testing is becoming increasingly helpful in pediatric practice to diagnose and monitor various lung conditions” (Jat and Agarwal). This device can quickly and accurately diagnose and monitor lung conditions such as asthma, COPD, and other respiratory inhibiting techniques. This is an important aspect since our onesie will be used as a means for an accessible way to monitor the lung function of infants, quickly and accurately reporting any abnormalities. The spirometry test is generally safe as it only requires the patient to use specific breathing techniques; however, the onesie will not use any mechanics that require usage from the patient’s oral cavity.

Other strengths of this method include that it can quickly and accurately diagnose and monitor lung conditions such as asthma, COPD, and other “obstructive and restrictive diseases” that affect the lungs (Jat and Agarwal). It can also be safely administered. Due to these strengths, this method is the most commonly used and available.

With the obvious strengths that this design contains, there are unavoidable drawbacks. First off, spirometry is only accessible in medical and respiratory therapy facilities. This means for the most part, spirometry will require medical professionals and consent to be able to use, which isn’t desirable when conducting research. The instructions and sequences of maneuvers the patient must conduct listed above may not seem intuitive to children of young ages, especially infants. Infants cannot understand simple commands thus it will be highly difficult for them to perform these breathing techniques listed above. If the patient fails to perform the specific breathing techniques, the spirometer cannot accurately assess their lung capacity and might even lead to overlooking or falsely predicting certain lung diseases in the patient. This is crucial since spirometry is often used as a diagnostic to assess the severity of lung

condition so the patient can be assigned to more niche PFTs. Overall the spirometer fails to meet most of the design requirements for our project. Although it is an effective method to monitor and assess breathing parameters, it can only be accessed through healthcare facilities. The spirometer measures lung capacity through assessing airflow from the oral cavity whereas the onesie must utilize chest compressions to determine lung capacity.

The shortcomings of this method are how it must be operated by a medical professional, the children/baby must be able to understand instructions, and that the results of the test can be affected by movement.

Something to consider for the onesie is that it can be used in a varying environments with both high equipment and minimal equipment thus making the onesie adaptable. Another is to not confine the movement of the infant too much to prevent distress.

Another application of the Spirometry test is to apply it at home. In the article *Spirometry and Plethysmography: Combining Diagnostic Power* analysed the success of an at home spirometry test. Having the procedure being able to do it at home saved cost and wait time. Clinical spirometry equipment costs around \$1500 and it takes a lot of resources to train staff and find time to do the readings (Halon). The readings alone can take up to 15-20 minutes to do to get a decent reading. And appointments usually take up to 30 days to process, by then the patient usually breaks it by then.

Having an at home spirometry test is supposed to cut down on costs and actually have the patient take the test. According to the article, "In-home testing was done 87% of the subjects had a better understanding of their diseases and 79% planned to continue using their in-home testing"(Halon).

What can be applied to our own device is to ensure that it will not need big amounts of energy and time to operate. The spirometry test is inconvenient in the first place because of the amount of resources it uses to function, having our device use less resources will make it easier to use and more convenient.

However, this method also has a couple of shortcomings. The study on this method wasn't conducted on younger people, which would be more valuable to this group and project. With this device coming solely from this study, it also is not a finalized product. It is still being tested.

Body Plethysmography

<https://my.clevelandclinic.org/health/diagnostics/15467-body-plethysmography-pulmonary-function-test>

Analysis:

Body plethysmography provides total lung capacity

Body plethysmography is a type of lung function test. You enter a see-through plastic box, where machines measure how much air goes into and out of your lungs when you breathe. Body plethysmography is safe, but deep breathing may make you feel lightheaded or dizzy. Your healthcare provider will contact you after a few days with your test results. It can also compare how your lungs function to how well your lungs should be working compared to any previous tests you've taken or to standards for your demographic. Measure the effect of chronic (long-lasting) lung diseases on your lung function. Detect early changes in your lung function that might require a change in your treatment. Determine whether substances in your home or work environment have damaged your lungs. Determine your ability to tolerate medical procedures, including surgery.

Types of body plethysmography include Pressure plethysmograph, which measures your lung volumes by measuring the changes in gas pressure in an airtight box. The box contains a fixed amount of air. A Spirometer is used to measure the volume in the area between your neck and abdomen (thorax). This PFT is very sensitive and can measure small changes in volume.

There are also Volume Plethysmographs, where it measures gas volume changes by measuring gas displacement in a clear, airtight plastic box. When the gas volume changes in your thorax, the surrounding air in the box exits through a hole in a wall connected to a spirometer. The spirometer measures the airflow. A volume plethysmograph is challenging, so only medical researchers use this test. Another name for a volume plethysmograph is an open-type plethysmograph.

Finally, a Pressure-Volume Plethysmograph. This test combines elements of a pressure plethysmograph and a volume plethysmograph. As you breathe, any volume changes in your thorax force the surrounding air in the box through a hole in the wall into a spirometer. Your healthcare provider will then add the measurements in pressure change or volume displacement together.

Cleveland Clinic. "Pulmonary Function Test (Plethysmography) | Cleveland Clinic." Cleveland Clinic, 2018, my.clevelandclinic.org/health/diagnostics/15467-body-plethysmography-pulmonary-function-test.

Strengths and Weaknesses

<https://www.sciencedirect.com/science/article/pii/S0954611111000552>

Body plethysmography allows for respiratory therapists to detect early signs and abnormalities in one's lungs. Body plethysmography is able to calculate the total lung capacity, airway resistance, and residual volume (the air that remains in the lungs after ma exhalation. It can accurately measure the lung volume and capacity of the patient, however, body plethysmography cannot analyze the chest circumference across the upper, middle, and lower chest at the top of inspiration and end of exhalation. Lung capacity cannot be used to accurately predict the chest circumference, as they do not have a direct correlation with each other.

Criée, C.P., et al. "Body Plethysmography – Its Principles and Clinical Use." *Respiratory Medicine*, vol. 105, no. 7, July 2011, pp. 959–971, <https://doi.org/10.1016/j.rmed.2011.02.006>.

Semester 2 Research